Aim: String Manipulation.

1.Write a java program that illustrate the usage of String classStringBuffer class methods.

CODE:

```
public class DemoStringMethod {
  public static void main(String args[]){
    String s1="Glacier M416";
    String s2="Glacier AKM";

    // 1. trim()
    System.out.println("1. Trimmed: '" + s1 + "'");

    // 2. length()
    System.out.println("2. Length: " + s1.length());

    // 3. toLowerCase()
    System.out.println("3. Lowercase: " + s2.toLowerCase());

    // 4. toUpperCase()
    System.out.println("4. Uppercase: " + s1.toUpperCase());

    // 5. equals()
    System.out.println("5. Equals: " + s1.equals(s2));

    // 6. equalsIgnoreCase()
    System.out.println("6. EqualsIgnoreCase: " + s1.equalsIgnoreCase(s2));

    // 7. charAt()
    System.out.println("7. Character at index 2: " + s2.charAt(2));

    // 8. substring(int)
    System.out.println("8. Substring from index 5: " + s1.substring(5));

    // 9. substring(int, int)
    System.out.println("9. Substring from 5 to 11: " + s2.substring(5,10));

    // 10. contains()
    System.out.println("10. Contains 'AKM': " + s2.contains("AKM"));

    // 11. startsWith()
    System.out.println("11. Starts with 'Glacier': " + s1.startsWith("Glacier"));

    // 12. endsWith()
    System.out.println("12. Ends with '416': " + s1.endsWith("416"));

    // 13. replace()
    System.out.println("13. Replace 'M' with '@': " + s1.replace('M', '@'));

    // 14. indexOf()
```

```java
        System.out.println("14. Index of 'M': " + s2.indexOf('M'));

        // 15. isEmpty()
        System.out.println("15. Is empty string actually empty? " + s1.isEmpty());
    }
}
```

```java
public class DemoStringBuffer{
    public static void main(String[] args) {
        StringBuffer s = new StringBuffer("DemonGreeds");

        // 1. append()
        s.append("!");
        System.out.println("1. After append: " + s);
        // 2. insert()
        s.insert(5, " ");
        System.out.println("2. After insert: " + s);
        // 3. replace()
        s.replace(6, 12, " Blood");
        System.out.println("3. After replace: " + s);
        // 4. delete()
        s.delete(0, 6);
        System.out.println("4. After delete: " + s);
        // 5. reverse()
        s.reverse();
        System.out.println("5. After reverse: " + s);
    }
}
```

OUTPUT:

```
1. Trimmed: 'Glacier M416'
2. Length: 12
3. Lowercase: glacier akm
4. Uppercase: GLACIER M416
5. Equals: false
6. EqualsIgnoreCase: false
7. Character at index 2: a
8. Substring from index 5: er M416
9. Substring from 5 to 11: er AK
10. Contains 'AKM': true
11. Starts with 'Glacier': true
12. Ends with '416': true
13. Replace 'M' with '@': Glacier @416
14. Index of 'M': 10
15. Is empty string actually empty? False
```

```
1. After append: DemonGreeds!
2. After insert: Demon Greeds!
3. After replace: Demon  Blood!
```

4. After delete: Blood!
5. After reverse: !doolB

2.Write a java program that illustrate the usage of StringTokenizer class methods.

CODE:

```
import java.util.StringTokenizer;
public class DemoString {
  public static void main(String args[]){
    String s="Once, I was a great, invincible warrior, and no-one dared to stand in my way.";
    StringTokenizer tokens = new StringTokenizer(s);
    System.out.println("Original Sentence: "+s);
    System.out.println("Total number of tokens present in a sentence: "+tokens.countTokens());
    System.out.println("Tokens:");
    while(tokens.hasMoreTokens()){
      String ss = tokens.nextToken();
      System.out.println(ss);
    }
  }
}
```

OUTPUT:

```
Original Sentence: Once, I was a great, invincible warrior, and no-one dared to stand in my way.
Total number of tokens present in a sentence: 15
Tokens:
Once,
I
was
a
great,
invincible
warrior,
and
no-one
dared
to
stand
in
my
way.
```